

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

Fläche unter der Glockenkurve

$$\text{Sei } J := \int_{-\infty}^{\infty} e^{-x^2} dx$$

Nicht integrieren wir über zwei unabhängige Variablen:

$$\begin{aligned} J^2 &= \int_{-\infty}^{\infty} e^{-x^2} dx \cdot \int_{-\infty}^{\infty} e^{-y^2} dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \end{aligned}$$

Wir führen Polarkoordinaten ein:

$$\begin{aligned} x &= r \cos(\theta) & \rightarrow \frac{dx}{dr} &= \cos(\theta), \quad \frac{dx}{d\theta} = -r \sin(\theta) \\ y &= r \sin(\theta) & \frac{dy}{d\theta} &= r \cos(\theta), \quad \frac{dy}{dr} = \sin(\theta) \end{aligned}$$

$$\begin{aligned} dx &= \frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta \\ dy &= \frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta \end{aligned}$$

$$\rightarrow \left. \begin{aligned} dx &= \cos(\theta) dr - r \sin(\theta) d\theta \\ dy &= r \cos(\theta) d\theta + \sin(\theta) dr \end{aligned} \right\} dy = r \frac{dx}{dr} d\theta$$

$\rightarrow$ Jacobimatrix beschreibt die Transformation

$$\text{Jac} = \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \Rightarrow |\det(\text{Jac})| = |r| = r$$

$$\Rightarrow dx dy = r dr d\theta$$

Mit den totalen Differenzialen:

$$dx = \frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta = \cos(\theta) dr - r \sin(\theta) d\theta$$

$$dy = \frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta = \sin(\theta) dr + r \cos(\theta) d\theta$$

$$\rightarrow dx dy = (\cos(\theta) dr - r \sin(\theta) d\theta) (\sin(\theta) dr + r \cos(\theta) d\theta)$$

$$= \sin(\theta) \cos(\theta) dr^2 + r \cos^2(\theta) dr d\theta$$

$$- r \sin^2(\theta) d\theta dr - r^2 \sin \theta \cos(\theta) d\theta^2$$

= ... Die Fläche wird mit dem Kreuzprodukt

der Vektoren $(\cos \theta, \sin \theta) dr$, $(-r \sin \theta, r \cos \theta) d\theta$ bestimmt.

$$\Rightarrow J^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

$$= \int_0^{2\pi} \cdot \frac{1}{2} \int_0^{\infty} e^{-u} du d\theta$$

$$= \int_0^{2\pi} \frac{1}{2} d\theta = \frac{1}{2} \theta \Big|_0^{2\pi} = \pi$$

$$\Rightarrow J = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

□

einfacher: über kartesische Koordinaten

Wiedrum ähnlich $I^2 = \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy$

Setze aber nun $x^2 + y^2 =: s$

$$\Rightarrow \frac{ds}{dx} = 2x \quad \text{z. } x=0 \Rightarrow s=y^2$$

$$x \rightarrow \infty \Rightarrow s \rightarrow \infty$$

$$\Rightarrow I^2 = \int_0^{\infty} \int_{y^2}^{\infty} e^{-s} \frac{ds}{2x} dy$$

$$\text{beachte: } x = \sqrt{s - y^2} \Rightarrow dx = \frac{ds}{2\sqrt{s - y^2}}$$

$$\begin{aligned}\Gamma^2 &= \int_0^\infty \int_{-\infty}^\infty e^{-s} \frac{ds}{2\sqrt{s-y^2}} dy \\ &= \int_0^\infty e^{-s} \left(\int_0^{\sqrt{s}} dy \right) ds \\ &= \int_0^\infty e^{-s} \sqrt{s} ds\end{aligned}$$

Es scheint am einfachsten, über die Polarkoordinaten zu gehen. Denn man müsste jetzt die Gamma-Funktion ins Spiel bringen:

$$\Gamma(n) := \int_0^\infty s^{n-1} e^{-s} ds$$

$$\rightarrow \Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right)$$

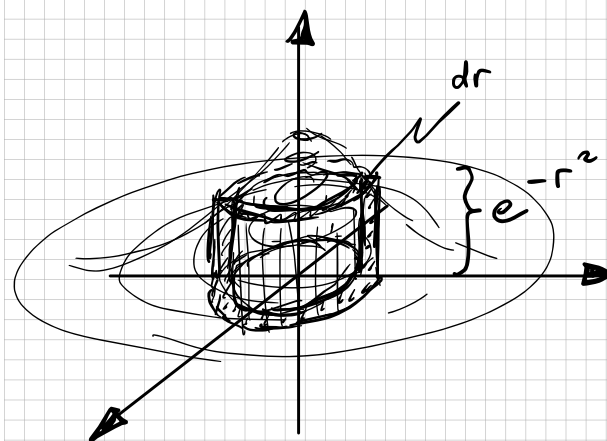
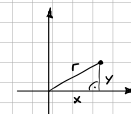
Approach from Sandhu (3 blue, 1 brown)

But also geometrical ...

$$f_1(x) = e^{-x^2}$$

$$f_2(x, y) = e^{-(x^2+y^2)} = e^{-r^2}$$

$$x^2 + y^2 = r^2$$



Volume

$$2\pi r \cdot e^{-r^2} \cdot dr$$

$$\int_0^\infty 2\pi r \cdot e^{-r^2} dr = \pi \int_0^\infty 2r e^{-r^2} dr = \pi \left(-e^{-r^2} \Big|_0^\infty \right) = \underline{\pi}$$

$$\text{Now } y=0 \rightarrow e^{-(x^2+y^2)} = e^{-x^2}.$$

$$\text{Set } \int_{-\infty}^\infty e^{-x^2} dx = \underline{C}.$$

Calculate $\int_{-\infty}^{\infty} C \cdot e^{-y^2} dy = C \int_{-\infty}^{\infty} e^{-y^2} dy = C^2$

$\Rightarrow C^2 = \pi \Rightarrow \underline{\underline{\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}}}$

Üb: Berechne $\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$

observe: $e^{-\frac{(x-\mu)^2}{2\sigma^2}} = e^{-\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)^2}$

$\rightarrow$ set $y := \frac{x-\mu}{\sqrt{2}\sigma} \Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{2}\sigma} \Leftrightarrow dx = \sqrt{2}\sigma dy$

$\Rightarrow \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-y^2} \cdot \sqrt{2}\sigma dy = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2} dy = \frac{1}{\sqrt{\pi}} \pi = 1$